Buzau Cosmin-Ionut

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Education

Technical University "Gheorghe Asachi" of Iași Oct 2024 – 2027 (Expected)
B.Sc. in Computer Science and Information Technology *Iași, România*

- **Performance scholarship** recipient; admitted among the **top applicants** to the program (2024).
- **Coursework:** Object-Oriented Programming (Advanced), Data Structures & Algorithms, Databases, Computer Architecture & Assembly, Operating Systems (Linux).

Experience

NIMA SOFTWARE SRL Aug 2026 – Sep 2026
AI Software Engineer (Project-based contract) — CargoPlanning *Remote*

- Rebuilt the company blog (Symfony 6.1) from 17 hardcoded Twig templates into a flat-file Markdown system rendered dynamically in 9 languages, with every existing URL preserved.
- Built an event-driven **email-to-article pipeline** for guest posts: an **AWS SES** receipt rule stores each raw email in **S3** and notifies an **SNS** topic, triggering a service that parses the MIME message (sender, body, Word/PDF attachments, images) into a structured draft.
- Integrated a **self-hosted LLM** (Qwen) that classifies each email (submission, request, spam) and checks topic, links and images; the author's text is kept verbatim and translated into all 9 languages, with no data sent to a third-party AI API. Output is written as Markdown and goes live only after an editor reviews and commits it.
- Hardened the pipeline against **prompt injection**: email content is isolated as untrusted data behind markers the sender cannot forge, and the model's output must match a fixed format before it is accepted; **790 adversarial test cases** confirm that even a successfully manipulated model cannot change the outcome.

Self-Employed Jan 2025 – Present
Full-Stack & AI Developer *Iași · Remote*

- Deliver and maintain 3 production websites for paying clients end-to-end — design, code, database, deployment and domains — each with an admin flow so owners update content without a developer.
- **Compact Drive** (compactdrive.ro) — driving school platform on Next.js + Supabase; technical SEO drives **150+ organic visits/month** from Google, and enrollment requests arrive instantly by email.
- **Boeru's Doner** (boerus-d.ro) — fast-food chain platform on Next.js + Tailwind with digital menu, promotions and a Leaflet map of branches.
- **Aeron Termosistem** (aerontermosistem.vercel.app) — construction site on Next.js, MongoDB and Cloudinary; client-managed gallery, leads saved to the DB.
- Design and ship AI automations for business clients (chat assistants, voice and document processing, smart search), e.g. turning photographed receipts into accountant-ready Excel data.

Projects

AI Trading Bot — Self-Retraining ML System | *Python, PyTorch, LightGBM, MQL5, AWS S3* GitHub

- Self-improving MetaTrader 5 bot built on multi-year price history and engineered market features; every night it backtests strategies on fresh data to generate labeled signals and retrains two competing models (LightGBM, then a PyTorch network), periodically tuned with Optuna.
- Connected to **two exit agents** that manage open trades: an **AI agent** that reads the Forex Factory calendar and closes positions exposed to high-impact news, and a **rule-based agent** that reviews each trade every 60s against algorithmic exit rules and holds it unless one triggers.
- Weekly backups to AWS S3; live monitoring via a Flask dashboard and daily Telegram reports.

AI Sales Agent — Microsoft Teams | *Python, FastAPI, Azure AD, Llama 3.1, Whisper* GitHub

- AI agent that answers sales staff's questions about products, prices and offers inside Microsoft Teams or on the web, by text or voice note (Whisper large-v3), connected to Microsoft 365 via Azure AD (OAuth2).
- Three-step product search robust to typos, synonyms and translated names — exact code match, fuzzy scoring (sentence-transformers), then an LLM double-check; asks one short question when a request is ambiguous.
- Turns a confirmed request into a ready-to-send PDF offer on the spot (ReportLab).

Projects (continued)

Floraly — AI Sales Agent for Flower Shops | *Next.js, TypeScript, Supabase, Claude API* *In development*

- AI agent that sits as a chat widget in the corner of a flower shop's website (and later on in-store tablets): it asks about the occasion, the recipient and the budget, then recommends matching bouquets.
- Bouquets live in a Supabase (PostgreSQL) catalog with price, flowers, colors, occasions and stock; the LLM queries it via tool calls filtered by budget and event, so it only suggests bouquets that are actually available.
- Each conversation is saved as a structured request (bouquet, occasion, budget, contact) in a Next.js dashboard, where the florist prepares the order or follows up with the customer.

Serverless Claude Chat on AWS Bedrock | *Python, Lambda, API Gateway, SAM, Next.js* GitHub

- Serverless chat API: a Python Lambda behind API Gateway calls Claude on Amazon Bedrock, with all infrastructure as code (AWS SAM), redeployable with one command.
- Least-privilege IAM role whose only permission is invoking the model; a Next.js 15 client streams the answer word-by-word as formatted Markdown with highlighted code.

Fără Agenții — Commission-Free Real Estate Marketplace | *Next.js, Supabase, Groq* Link | GitHub

- Full-stack marketplace where owners in Iași sell or rent directly to buyers, with no agents or commissions: search and filters by neighborhood, property type and price, an owner dashboard with listing statistics, and an admin panel to moderate listings, users and reports.
- PostgreSQL schema of 7 tables with **Row Level Security on every table** — users can only access their own listings, favorites and conversations, admins get separate policies; database triggers create profiles on signup, and buyer–owner chat runs live on Supabase Realtime.
- **AI assistant** (LangChain + Llama 3.3 70B on Groq) that recommends listings in Romanian: it retrieves the 6 best-matching active listings via diacritic-insensitive similarity search and answers only from them.

EmotionCamera — Real-Time Emotion Recognition | *Next.js, FastAPI, OpenCV, TensorFlow* GitHub

- Full-stack computer-vision web app that detects faces in a live webcam stream and classifies 7 facial expressions in real time, overlaying the emotion, its confidence and the face box directly on the video.
- Frames are downscaled on a hidden canvas in the browser before upload, cutting payload size so the FastAPI backend keeps up at about 30 FPS; OpenCV finds faces and a FER (TensorFlow) CNN classifies them.
- Split deployment: containerized Python backend on Railway (Docker), Next.js 14 frontend on Vercel.

Trapped in the Server — 2D Java Game, No Engine | *Java, Java2D / AWT, Tiled, SQLite* GitHub

- Pure-Java game with a custom 60 FPS loop, three levels and a boss fight, built on OOP design patterns (Singleton, State, Abstract Factory) with SQLite save/load; two-person university project.

Technical Skills

Languages: Python, TypeScript, JavaScript, PHP, Java, C, C++

Frameworks: Next.js, React, FastAPI, Flask, Symfony, Tailwind CSS, LangChain

AI & ML: LLM integration, AI agents, RAG, Claude, Whisper, PyTorch, LightGBM

Cloud: AWS (Lambda, S3, SES, SNS, SAM, Bedrock), Docker, Git, Linux, Vercel

Databases: PostgreSQL, Supabase, MongoDB, SQLite

Spoken Languages: Romanian (native), English (B2)

Open to Full Stack / Software Engineer / AI Engineer roles

full-time, part-time, internship or B2B collaboration · Iași or remote